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CHARCATERISTICS OF OPERATIONAL AMPLIFIERS

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Nome della conferenza o seminario

Different types of Operational amplifiers



- Voltage mode operational amplifier
- Current feedback
- Norton operational amplifier

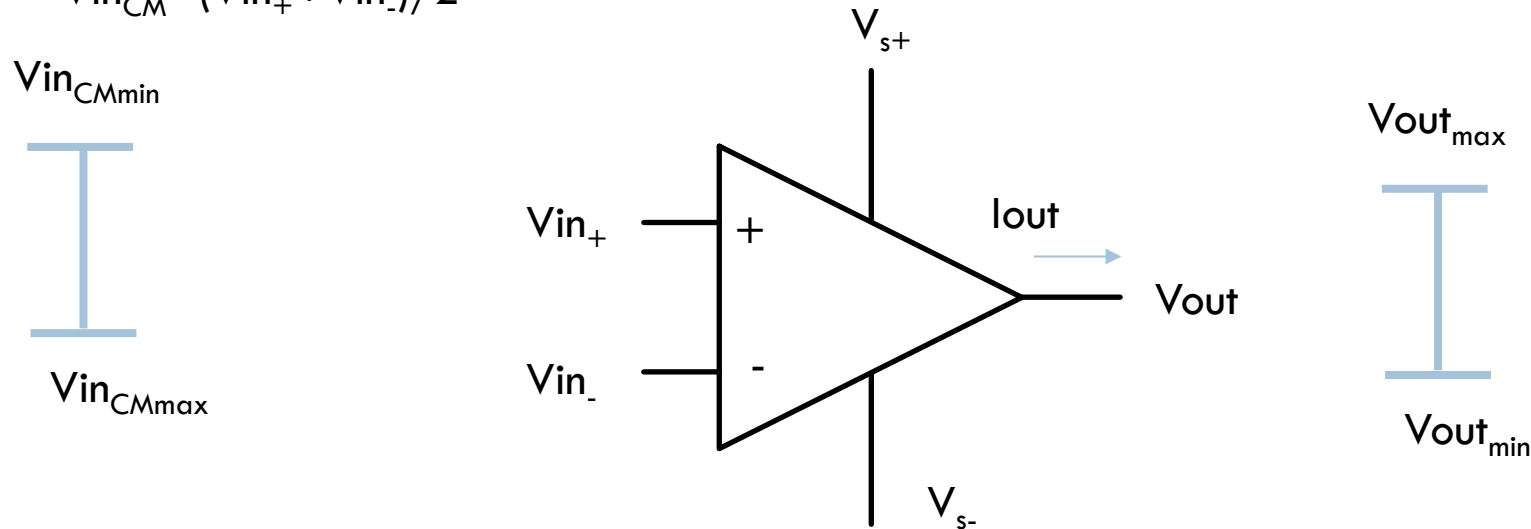
OPAMP general Characteristics

- Polarization and/or large signal
 - Input common mode range $ICMR$
 - Output voltage swing $V_{out_{max}}$ $V_{out_{min}}$
 - Maximum output current
 - Input bias current
 - Slew rate SR
- AC characteristics
 - Input impedance
 - DC Gain $A_v(0)$
 - *Output impedance*
 - Gain band product
 - $CMRR$
 - $PSRR$
 - Input equivalent noise
- *Mismatch non idealities*
 - Input offset V_{offset}
 - *Input current mismatch*

Polarization and/or large signal

- Input common mode range ICMR
- Output voltage swing $V_{out_{max}}$ $V_{out_{min}}$

$$V_{in_{CM}} = (V_{in_{+}} + V_{in_{-}}) / 2$$



Se $V_{in_{CM}}$ è nel range $(V_{in_{CMmin}}, V_{in_{CMmax}})$, se V_{out} è nel range $(V_{out_{min}}, V_{out_{max}})$ e se la corrente di uscita I_{out} è nel range $(I_{out_{min}}, I_{out_{max}})$ dove $I_{out_{min}}$ è la massima corrente entrante (Massimo sinking) e $I_{out_{max}}$ è la massima corrente uscente (Massimo sourcing).



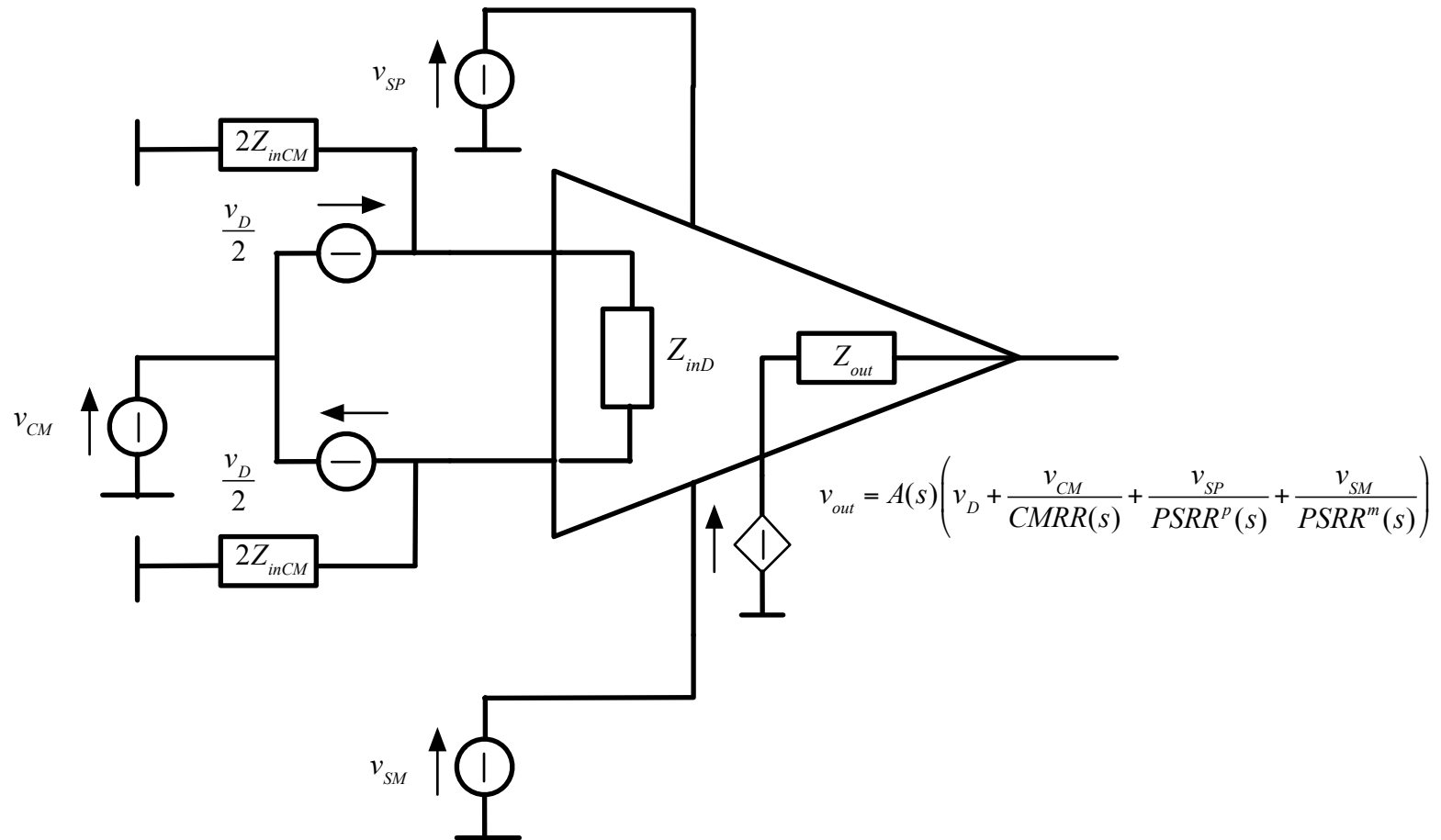
L'amplificatore operazionale è polarizzato correttamente e può essere studiato con il suo modello di piccolo segnale.

AC linear model Small Signal



- Input impedance differential
- Output impedance
- Gain transfer function (GBWP)
- CMRR
- PSRR

Scheme

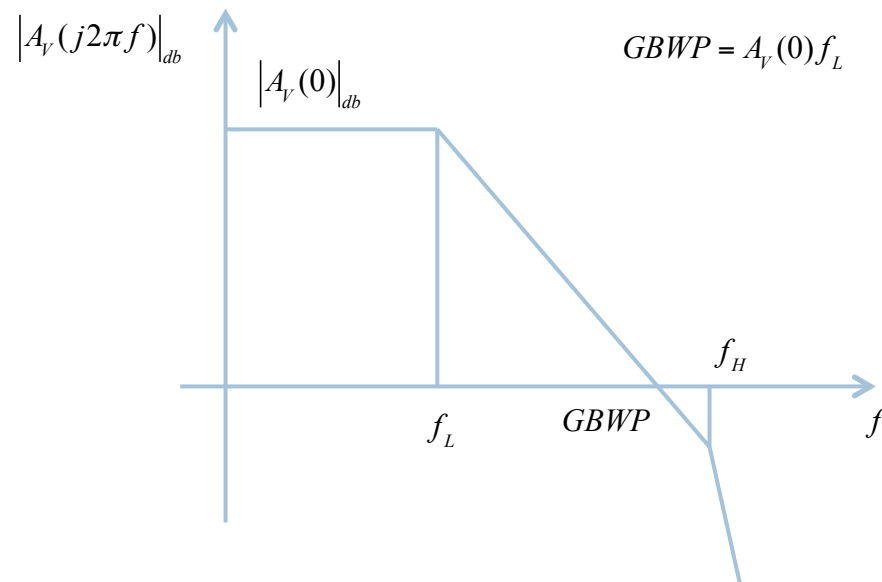


Gain transfer function

□ Transfer function

$$\frac{v_{OUT}}{v_D} = A_V(s)$$

$$A_V(s) = \frac{A_V(0)}{(1 + s\tau_L)(1 + s\tau_H)}$$



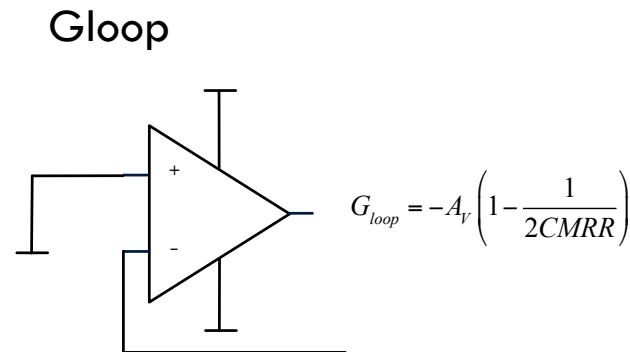
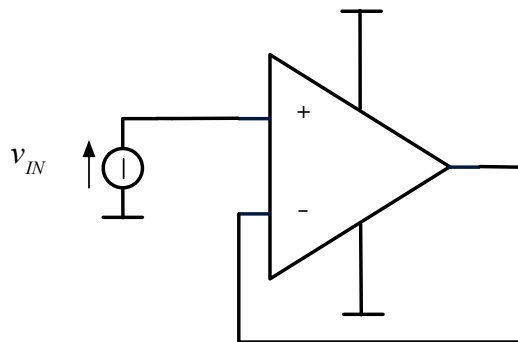
Example CMRR

- Considering CMRR in the transfer function

$$\frac{v_{OUT}}{v_{IN}} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} \quad v_{in} - v_{out} + \frac{v_{in} + v_{out}}{2CMRR} = 0 \quad \Rightarrow \quad \frac{v_{out}}{v_{in}} = \frac{1 + \frac{1}{2CMRR}}{1 - \frac{1}{2CMRR}}$$

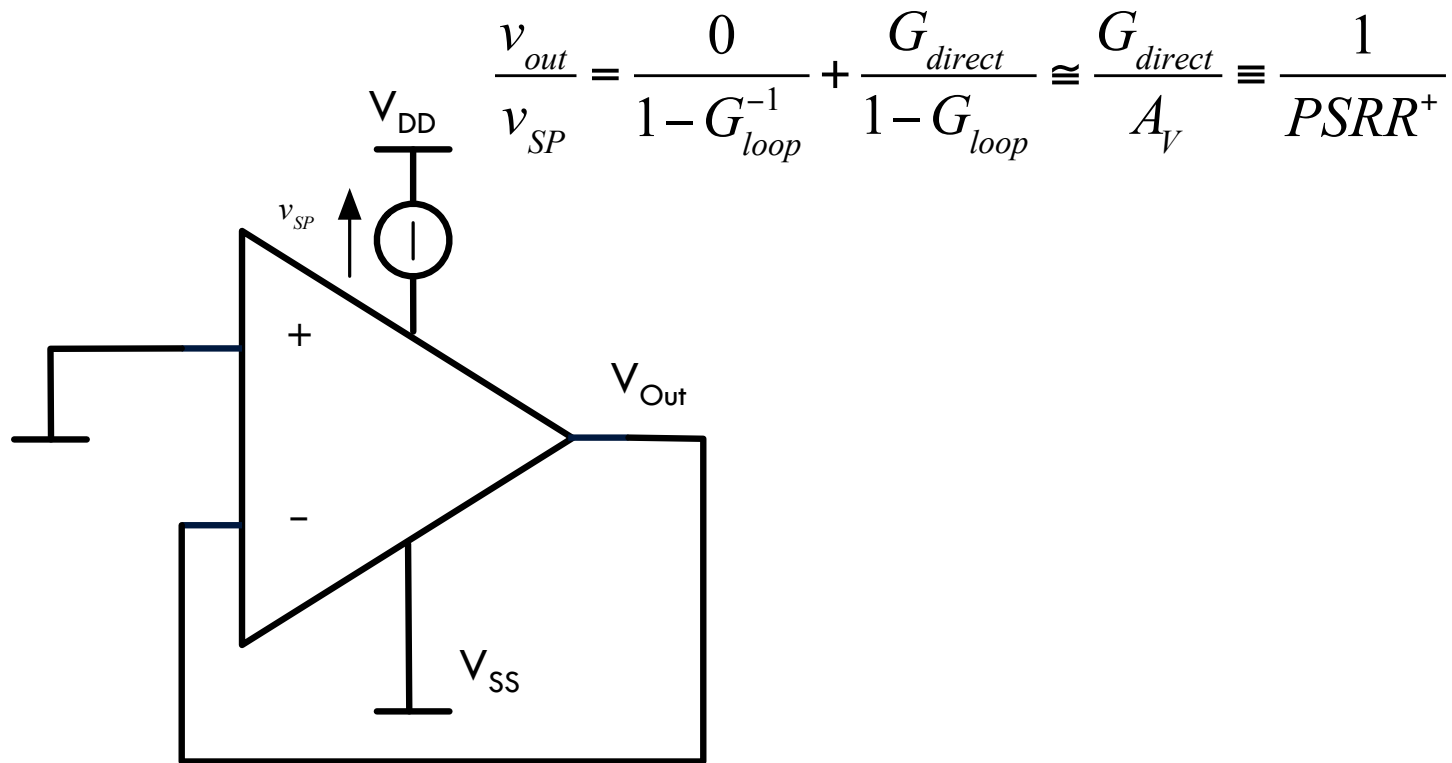
$$\frac{v_{OUT}}{v_{IN}} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} = \frac{A_V \left(1 + \frac{1}{2CMRR} \right)}{1 + A_V \left(1 - \frac{1}{2CMRR} \right)}$$

$$\frac{v_{out} - v_{in}}{v_{in}} = \frac{1 + \frac{1}{2CMRR}}{1 - \frac{1}{2CMRR}} - 1 = \frac{\frac{1}{CMRR}}{1 - \frac{1}{2CMRR}} \approx \frac{1}{CMRR}$$



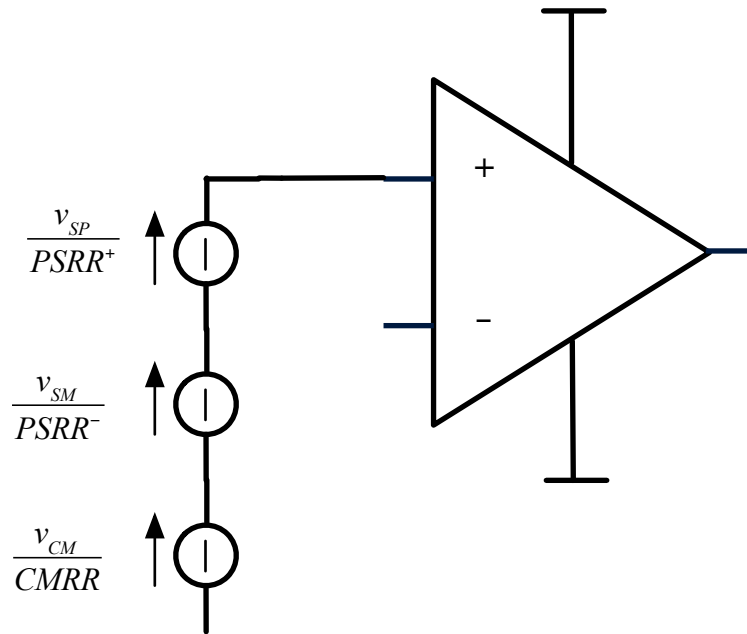
Example PSRR

- Signal noise on vdd



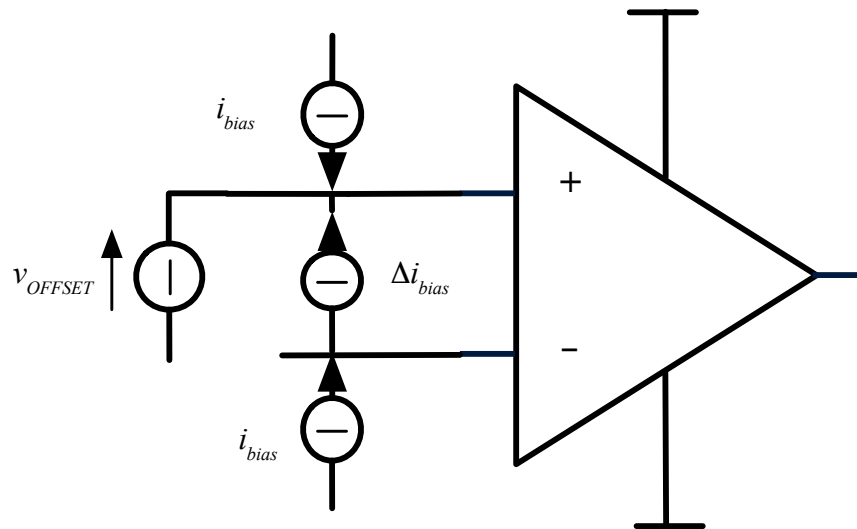
Equivalent circuit

□ Equivalent circuit



Effetto di offset e bias

□ Circuito equivalente



Limitazioni di grande segnale

- Limitazione dovuta alla SR considerando un segnale sinusoidale di uscita di ampiezza A e frequenza ω

$$A\omega < SR$$

- Limitazione dovuta alla I_{out} sulla massima derivata di uscita

$$A\omega < \frac{I_{out}}{C_L}$$